

A NON-RESOLVABLE $(7 \times 15, 35)$ -TRIPLE ARRAY

ABSTRACT. Gordeev and Öhman ask, in Problem 58 of [1], for a non-resolvable $(7 \times 15, 35)$ -triple array. We exhibit one. The witness is printed in full, and its non-resolvability is decided by a single multiset a reader can read off the array: the 35 symbols realise 31 distinct row-triples, with maximum multiplicity 2, whereas a resolution needs five symbols sharing one row-triple.

1. THE PROBLEM

We follow [1, Definition 6]. For an $r \times c$ array T on a symbol set of size v , write R_i for the set of symbols occurring in row i and C_j for the set of symbols occurring in column j .

Definition 1. T is an $(r \times c, v)$ -triple array if

- (T1) no symbol occurs more than once in a row and no symbol occurs more than once in a column (so T is *binary* in rows and in columns);
- (T2) every symbol occurs exactly $e = rc/v$ times;
- (T3) $|R_i \cap C_j| = \lambda_{rc}$ for all i, j ;
- (T4) $|R_i \cap R_s| = \lambda_{rr}$ for all $i \neq s$;
- (T5) $|C_j \cap C_t| = \lambda_{cc}$ for all $j \neq t$.

For $(r, c, v) = (7, 15, 35)$ the parameters are forced: $e = rc/v = 3$, $\lambda_{rc} = e = 3$, $\lambda_{rr} = c(e - 1)/(r - 1) = 5$ and $\lambda_{cc} = r(e - 1)/(c - 1) = 1$. In the notation of [1] the object below is a $\text{TA}(35, 3, 5, 1, 3 : 7 \times 15)$.

Definition 2 ([1, Definition 21]). An $(r \times c, v)$ -triple array is *resolvable* if its v symbols can be partitioned into r groups of size $k = c/e$ such that all symbols of a group occur in the same set of rows, and every column set C_j contains precisely one symbol of each group.

Here $k = c/e = 5$ is an integer and $rk = 7 \cdot 5 = 35 = v$, so the numerical conditions that Definition 2 imposes on the parameters are met; these are necessary and not sufficient, but resolvable arrays on this cell are known [1]. Non-resolvability here is therefore a genuine property and not an integrality obstruction. The same paper then asks, in Section 9.2 (“Open questions”):

Problem 58. Construct a non-resolvable $(7 \times 15, 35)$ -triple array.

2020 *Mathematics Subject Classification.* 05B05, 05B30, 62K10.

Key words and phrases. triple array, row-column design, resolvable design, non-resolvable design.

The preceding paragraph of [1] supplies the quantifier: “a triple array does not have to be resolvable even when its parameter set is admissible for resolvable triple arrays. Despite our full enumeration of resolvable $(7 \times 15, 35)$ -triple arrays, it is therefore still an open question whether there are further, non-resolvable triple arrays on these parameters.” Definitions 1 and 2 are reproduced from [1] and everything below is relative to them as reproduced: an array satisfying Definition 1 and failing Definition 2 settles the problem as stated there, and no other reading of “resolvable” is treated.

2. THE ARRAY

Rows $1, \dots, 7$ downwards, columns $1, \dots, 15$ rightwards, symbols $1, \dots, 35$:

R_1	1	10	18	20	19	27	15	12	5	17	22	4	35	33	34
R_2	2	12	8	22	24	23	20	30	31	3	34	26	13	7	18
R_3	3	8	16	19	15	10	31	5	28	32	6	34	29	24	9
$T = R_4$	4	1	14	3	25	28	30	16	13	33	23	21	18	10	32
R_5	5	13	15	21	26	20	11	27	25	29	33	8	6	14	7
R_6	6	9	17	14	1	2	32	24	22	26	11	28	30	35	27
R_7	7	11	2	9	23	29	4	21	17	12	16	35	19	31	25

Theorem 3. *T is a $(7 \times 15, 35)$ -triple array in the sense of Definition 1, and T is not resolvable in the sense of Definition 2. Hence Problem 58 of [1], as reproduced above, has an affirmative answer.*

That T satisfies Definition 1 is a finite check on the printed array: all 105 cells carry symbols from $\{1, \dots, 35\}$, each of the 35 symbols exactly three times; each row holds 15 distinct symbols and each column 7; $|R_i \cap C_j| = 3$ on all 105 row-column pairs; $|R_i \cap R_s| = 5$ on all 21 row pairs; and $|C_j \cap C_t| = 1$ on all 105 column pairs. The array is non-trivial, $\max(r, c) = 15 < v = 35 < rc = 105$.

Non-resolvability. For a symbol s let

$$\text{rs}(s) = \{i : s \in R_i\}$$

be its *row-set*. By (T1) and (T2) each symbol occurs three times in three distinct rows, so $\text{rs}(s)$ is a 3-subset of $\{1, \dots, 7\}$. Reading them off T :

$$\begin{aligned} \text{rs}(1) &= 146 & \text{rs}(2) &= 267 & \text{rs}(3) &= 234 & \text{rs}(4) &= 147 & \text{rs}(5) &= 135 & \text{rs}(6) &= 356 \\ \text{rs}(7) &= 257 & \text{rs}(8) &= 235 & \text{rs}(9) &= 367 & \text{rs}(10) &= 134 & \text{rs}(11) &= 567 & \text{rs}(12) &= 127 \\ \text{rs}(13) &= 245 & \text{rs}(14) &= 456 & \text{rs}(15) &= 135 & \text{rs}(16) &= 347 & \text{rs}(17) &= 167 & \text{rs}(18) &= 124 \\ \text{rs}(19) &= 137 & \text{rs}(20) &= 125 & \text{rs}(21) &= 457 & \text{rs}(22) &= 126 & \text{rs}(23) &= 247 & \text{rs}(24) &= 236 \\ \text{rs}(25) &= 457 & \text{rs}(26) &= 256 & \text{rs}(27) &= 156 & \text{rs}(28) &= 346 & \text{rs}(29) &= 357 & \text{rs}(30) &= 246 \\ \text{rs}(31) &= 237 & \text{rs}(32) &= 346 & \text{rs}(33) &= 145 & \text{rs}(34) &= 123 & \text{rs}(35) &= 167 \end{aligned}$$

(here 146 abbreviates $\{1, 4, 6\}$).

Proof of Theorem 3. Suppose T were resolvable, with groups $G_1, \dots, G_7$ as in Definition 2. All $k = 5$ symbols of a group occur in the same rows, so

each G_i is contained in one fibre of rs . Consequently every fibre of rs is a disjoint union of groups, and in particular

$$(1) \quad 5 \mid |\text{rs}^{-1}(A)| \quad \text{for every } A \subseteq \{1, \dots, 7\}.$$

The list above takes exactly 31 distinct values, and the only coincidences are

$$\begin{aligned} \text{rs}(5) = \text{rs}(15) = 135, & \quad \text{rs}(21) = \text{rs}(25) = 457, \\ \text{rs}(28) = \text{rs}(32) = 346, & \quad \text{rs}(17) = \text{rs}(35) = 167. \end{aligned}$$

So the multiset of fibre sizes is $\{1^{27}, 2^4\}$, of total $27 + 8 = 35$, and its maximum is 2. As $2 < 5$, (1) fails — there is not even one set of five symbols with a common row-set, so no group can be formed at all. This contradiction proves that T is not resolvable. $\square$

3. VERIFICATION

The array above is the only input needed. Everything asserted about it — the five clauses of Definition 1, the row-set list and its fibre-size multiset $\{1^{27}, 2^4\}$, and the failure of Definition 2 — was recomputed in exact integer arithmetic by a standard-library Python program that reads the printed array and nothing else. The program (`verify.py`) and a recorded run of it (`verify.output.txt`) accompany this paper; the run also reports further checks on the array that the argument above does not use. It checks nothing bibliographic: the wording of Problem 58 and of the two definitions reproduced above is taken from [1] and is confirmed by no computation here. Neither file is needed to redo the work: the deciding fact, that no five symbols share a row-set, can be read off the row-set list by hand.

REFERENCES

- [1] A. Gordeev and L.-D. Öhman, *Resolvable triple arrays*, Electron. J. Combin. **33** (2026), no. 2, Paper No. 2.35. doi:10.37236/14977; arXiv:2512.08681v2.